

Protection by *Ziziphora clinopoides* of acetic acid-induced toxic bowel inflammation through reduction of cellular lipid peroxidation and myeloperoxidase activity

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Inflammatory bowel disease (IBD) is a chronic condition of the intestine with unknown etiology involving multiple immune, genetic and environmental factors. We were interested in examining the effect of a total extract from *Ziziphora clinopoides*, an Iranian folk herbal medicine, in the prevention and control of experimental mouse IBD. *Z. clinopoides* was administered (75, 150, 300 mg/kg) through drinking water to mice, which dispensed a toxic dose of acetic acid intrarectally. Prednisolone was used as the standard drug for comparison. Biochemical, macroscopic and microscopic examinations of the colon were performed. Biochemical evaluation of the inflamed colon was carried out using assays of myeloperoxidase (MPO) activity and thiobarbituric acid reacting substances (TBARS) as indicators of free radical activity and cellular lipid peroxidation.

Results indicated that the activity of MPO and lipid peroxidation products (TBARS) increased in acetic acid-treated groups, while recovered by pretreatment of

animals with *Z. clinopoides* (75–300 mg/kg) and prednisolone. All doses of *Z. clinopoides* and prednisolone-treated groups showed significant lower score values of macroscopic and microscopic characters when compared to the acetic acid-treated group. The beneficial effect of *Z. clinopoides* (300 mg/kg) was comparable to that of prednisolone.

It is concluded that *Z. clinopoides* inhibits acetic acid toxic reactions in the mouse bowel through inhibition of cellular oxidative stress. Proper clinical investigation should be carried out to confirm the same activity in human. *Human & Experimental Toxicology* (2006) 25, 325–332

Key words: acetic acid; cells lipid peroxidation; inflammatory bowel disease; mouse; myeloperoxidase; toxicity; oxidative stress; *Ziziphora clinopoides*

Introduction

Inflammatory bowel disease (IBD) comprises those conditions characterized by a tendency for chronic or relapsing immune activation and inflammation within the gastrointestinal tract. The intestinal inflammation is histologically characterized by

infiltration of polymorphonuclear leukocytes, monocytes, and macrophages. They are activated by various mediators, including prostaglandins, leukotrienes, platelet activating factor, and cytokines, to synthesize and liberate toxic reactive oxygen metabolites.^{1,2} Crohn's disease (CD) and ulcerative colitis (UC) are two major forms of IBD. Despite extensive investigation, the pathophysiology of human IBD is still not completely understood. Genetic and immunological factors, infective agents, smoking, medications and pathological factors have been reported to be involved in IBD.^{3,4}

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Oxidative stress as a well known mechanism of cell toxic response is believed to have role in the pathogenesis of intestinal damage.⁵ Considerable imbalance in total antioxidant power (TAP) (ie, increased oxidative stress and decreased antioxidant protection) in colonic mucosal biopsies has been reported.^{2,6,7} If there are too many free radicals or too few antioxidants for protection, a condition of oxidative stress develops, which may cause chronic and permanent damage to cells. One of the markers of oxidative stress is increased cellular lipid peroxidation, which is determined by measuring thiobarbituric acid reactive substances (TBARS).⁸ Previous studies showed the existence of oxidative stress in biological fluids, such as saliva,⁹ and plasma,^{5,7} of subjects with IBD. The effectiveness of various therapies in IBD has been suggested to be in relation to their antioxidant actions.^{10,12} Sulfasalazine and its metabolites,¹³ and 5-amino salicylic acid,^{12,14} used in the treatment of IBD are highly effective reactive oxygen metabolite scavengers. Neutrophils play a predominant role in inflammatory and immune reactions in IBD.¹⁵ Myeloperoxidase (MPO), which is mainly found in neutrophils, catalyses the oxidation of electron donor (eg, halides) by hydrogen peroxide. It has been reported that there is an increased level of MPO in the acetic acid mouse model of colitis.¹⁵ Administration of a toxic dose of acetic acid intrarectally induces bowel inflammation with a similar profile to human intestinal inflammation.¹⁶

Many herbal concoctions are thought to be effective in chronic inflammatory conditions. Labiatae are generally known for their various effects, such as analgesic and anti-inflammatory activity,¹⁷ antioxidant,¹⁸ hepatoprotective,¹⁹ and hypoglycemic action.²⁰ The genus *Ziziphora* L. belongs to the family Labiatae, consisting four species (*Z. clinopoides* LAM., *Z. capitata* L., *Z. persica* BUNGE., and *Z. tenuior* L.) is widespread in Iran. *Z. clinopoides* LAM. with the common Persian name of 'kakuti' comprised nine subspecies native to Iran. *Z. clinopoides* as an endemic subspecies, grows wildy in Iran and also Afghanistan, Iraq, and Tallish.²¹ In Iranian and Turkish folk medicine, *Ziziphora* species have been used as infusion for various purposes, such as sedative, stomach-ache and carminative.²² In Iranian folklore, the dried aerial parts of this plant have frequently been used as culinary and also in cold and cough treatments.²³ Regarding the above-mentioned evidence, we hypothesized that *Z. clinopoides* may alleviate IBD. In the present study, we demonstrated the status of oxidative stress by investigating bowel TBARS and

MPO activity and histopathological changes in mice.

Methods

Chemical

2-Thiobarbituric acid (TBA), 1,1,3,3-tetra ethoxypropan, trichloroacetic acid (TCA), *n*-butanol, hexadecyl trimethyl ammonium bromide (HETAB), EDTA, *O*-dianisidine hydrochloride, hydrogen peroxide, acetic acid, prednisolone, phosphate buffer, paraffin, hematoxyline, and eosin purchased from Merck Chemical Co. (Germany) were used in this study.

Plant material and extraction

The aerial parts (including flower, stalk and leaf) of *Z. clinopoides* were collected during its flowering stage in July 2004 from Tabriz (East Azerbaijan Province, Iran), identified, and stored in the Department of Botany of the Research Institute of Forests and Rangelands (TARI), Tehran. After drying and mincing the aerial parts of the plant, 25 g of the powder was used to obtain methanolic extracts by the percolation method. The methanolic extract was filtered and distilled at low temperature and vacuum (concentrated extract). Then, the concentrated extract was rinsed using chloroform and ether until the last extract became colorless. The chloroform solutions were separately concentrated and then collected to tally. Finally, 4-g of pure extract was obtained from the primary powder and the residue, which is water soluble, was used for administration to mice.

Animals

NMRI albino mice, weighing between 20 and 30 g, were used for the study. Animals were maintained under standard conditions of temperature ($23 \pm 1^\circ\text{C}$), relative humidity ($55 \pm 10\%$), and 12/12 hours light/dark cycle, and fed with a standard pellet diet and water *ad libitum*. They were housed in standard polypropylene cages with a wire mesh top. All studies were carried out using six mice/group. All ethical themes of the studies on animals were considered carefully and the experimental protocol was approved by the Ethic Committee of TUMS.

Induction of colitis and treatments

The protocol of the study, including doses and duration of treatment and groups, were designed according to previous studies.^{24,25} The study comprised six different groups as follows: normal or untreated animals – received no treatment; control animals – received 0.1 mL of 6% acetic acid

solution (once, intrarectally); *Z. clinopoides*-treated animals – received 7 days pretreatment with *Z. clinopoides* (75, 150, 300 mg/kg in drinking water) and 0.1 mL of 6% acetic acid solution; prednisolone-treated group – received prednisolone (1.14 mg/kg, for 3 days) and acetic acid (0.1 mL, 6% solution, once, intrarectally). Acetic acid was administered intrarectally on the eighth day and drug treatment was continued until the tenth day. Prednisolone as standard drug was started on the day of acetic acid administration and given orally as a suspension containing 0.5% of sodium CMC.

For induction of colitis, overnight fasted animals were anaesthetized using pentobarbital sodium (55 mg/kg, i.p.) and then 0.1 mL of 6% acetic acid solution was instilled into the rectum of the mouse. Animals were allowed to hang in air by holding their tails for 1–2 min to prevent spillage of the solution from the rectum. After 48 hours, the animals were sacrificed by cervical dislocation and dissected open to remove the colon. A long piece of colon (5 cm) was flushed gently with saline, cut open and scored for hyperemia and inflammation based on the macroscopic features by a histopathologist. For examination of microscopic features, tissues were fixed in 10% formalin saline and examined under microscope by a histopathologist. The histopathologist was completely blind to the treated samples. Biochemical evaluation of the colon inflammation was performed using assay of MPO activity and TBARS concentration.

Assay of colon macroscopic characters

The colon (5 cm long) was examined for hyperemia and lesions with inflammation and then scored according to the percentage of the 5-cm colon that was affected. The macroscopic features were scored as follow:

Percentage area affected	Score
0	0
1–5	1
5–10	2
10–25	3
25–50	4
50–75	5
75–100	6

Assay of colon microscopic (histologic) characters

To process for microscopic studies, 5 µm thick paraffin sections were stained in hematoxyline and eosin. The stained sections were examined for inflammatory changes, such as edema, granulated tissues, ulceration or thickening of mucosa, inflam-

mation, existence of polymorphonuclears and mononuclears, necrosis, disrupted architecture of the crypt, and narrow lumen. All sections were recorded by a histopathologist and a sign score, between 0 and 3 for each sign and a total of 24, were used to determine the severity of colon inflammation.

Assay of colon MPO activity

To measure MPO activity, colonic samples were minced on ice and homogenized in 10 mL of ice-cold 50 mM potassium phosphate buffer (pH 6.0), containing 0.5% hexadecyltrimethyl ammonium bromide (HETAB) and 10 mM EDTA. The homogenates were then sonicated and centrifuged for 20 min at 12 000 × *g*. MPO activity was measured spectrophotometrically as follows. The supernatant (0.1 mL) was combined with 2.9 mL of 50 mM phosphate buffer containing 0.167 mg/mL *O*-dianisidine hydrochloride and 0.0005% H₂O₂. The change in absorbance was measured spectrophotometrically (Shimadzu 160A UV-VIS spectrophotometer) at 460 nm. MPO activity (1 U) was defined as the change in absorbance per minute at room temperature. MPO activity (U/g) = $X/\text{weight of the piece of tissue taken}$, where $X = 10 \times \text{change in absorbance per min/volume of supernatant taken in the final reaction}$.²⁴

Assay of colon TBARS concentration

Malonedialdehyde (MDA) is the main end-product of the oxidation of polyunsaturated fatty acids and its concentration and other lipid peroxides in the medium is an established measure of the extent of lipid peroxidation. In this test, the reaction of TBA with lipid peroxide products produces a complex which is determined spectrophotometrically and lipid peroxidation in samples are assessed in terms of TBARS produced. Briefly, the colonic samples were homogenized in buffered saline (1:5) and then 800 µL of TCA (28% w/v) was added to 400 µL of this mixture and centrifuged in 3000 × *g* for 30 min. Then, the supernatant (600 µL) was added to 150 µL of TBA (1% w/v). The mixture was incubated for 15 min in a boiling water bath and then 4 mL *n*-butanol was added, the solution was centrifuged, cooled and absorption of the supernatant was recorded in 532 nm by UV-160-A Shimadzu double beam spectrophotometer (Japan). The calibration curve of a 1,1,3,3-tetraethoxypropan standard solutions was used to determine the concentrations of TBA-MDA adducts in samples. TBARS level was presented as µmol/g of colon.²⁶

Acute toxicity test

In order to determine the acute toxicity (LD50) of *Z. clinopoides*, doses of 10, 100, 1000 and 2000 mg/kg per day were gavaged to male Sprague–Dawley rats. The animals were observed for 48 hours and any mortality was recorded at the end of this period.

Statistical

Values are reported as mean $\pm$ SEM. Statistical significance between groups was computed by analysis of variance (ANOVA) and Tukey multiple comparison post-hoc tests. *P* values <0.05 were considered significant.

Results

Colon macroscopic and microscopic changes in control and *Z. clinopoides*-treated groups are shown in Tables 1 and 2, respectively. Intrarectal instillation of acetic acid caused significant inflammatory reactions as indicated by macroscopic and microscopic changes (Tables 1 and 2). *Z. clinopoides* (75–300 mg/kg)- and prednisolone-treated groups showed significantly lower score values for macroscopic and microscopic characters when compared to the control group. Positive effects of *Z. clinopoides* (300 mg/kg) in acetic acid-induced colon macroscopic and microscopic alterations were comparable to that of prednisolone.

The changes in the extent of lipid peroxidation in the bowel homogenate of treated animals are shown in Figure 1. The TBARS level significantly increased in the control group in comparison to the normal group (5.11 ± 0.50 versus 3.14 ± 0.31 $\mu\text{mol/g}$ colon). The TBARS level significantly decreased in *Z. clinopoides* (75–300)- and prednisolone-treated groups. The mean percentage of decreases of TBARS in *Z. clinopoides* (75, 150, 300) and prednisolone-treated groups were (7.9, 16.35, 33.9, and 35.6), respectively.

Table 1 Evaluation based on macroscopic features in acetic acid-induced colitis

Group	Mean of macroscopic scores $\pm$ SEM
Control	4.12 ± 0.48
Prednisolone	1.39 ± 0.11^a
<i>Z. clinopoides</i> 75	3.01 ± 0.42^a
<i>Z. clinopoides</i> 150	2.28 ± 0.37^a
<i>Z. clinopoides</i> 300	$1.31 \pm 0.12^{a,b}$

Observations were recorded by a histopathologist and scored as described in the methods section. Each value represents the mean of macroscopic scores $\pm$ SEM of six animals in each group.

^aSignificant (*P* <0.01) decrease in macroscopic score values compared to control.

^bThe difference in macroscopic score values of *Z. clinopoides*-treated and prednisolone-treated groups is not significant.

Table 2 Evaluation based on microscopic features in acetic acid-induced colitis

Group	Mean of microscopic scores $\pm$ SEM
Control	20.81 ± 0.45
Prednisolone	7.72 ± 0.53^a
<i>Z. clinopoides</i> 75	16.12 ± 0.72^a
<i>Z. clinopoides</i> 150	11.22 ± 0.59^a
<i>Z. clinopoides</i> 300	$7.21 \pm 0.46^{a,b}$

All histological findings were recorded by a histopathologist and scored as described in the methods section. Data are mean $\pm$ SEM of six observations.

^aSignificant (*P* <0.01) decrease in macroscopic score values compared to control.

^bThe difference in macroscopic score values of *Z. clinopoides*-treated and prednisolone-treated groups is not significant.

The changes in the activity of MPO in bowel homogenates of treated animals are shown in Figure 2. The MPO activity of the control group showed a significant increase in comparison to the normal group (4.76 ± 0.15 versus 2.89 ± 0.17 U/g colon). The MPO activity decreased in *Ziziphora* (75–300)- and prednisolone-treated groups compared to the control group. The mean percentages of decreases of MPO activity in *Z. clinopoides* (75, 150, 300)- and prednisolone-treated groups were (15, 26.5, 55.1, and 56.1), respectively.

Discussion

The present results showed that *Z. clinopoides* has a marked potential to suppress colitis in mice, as indicated by macroscopic, microscopic and toxicity biochemical evaluations. Biochemical assays well indicated that administration of *Z. clinopoides*

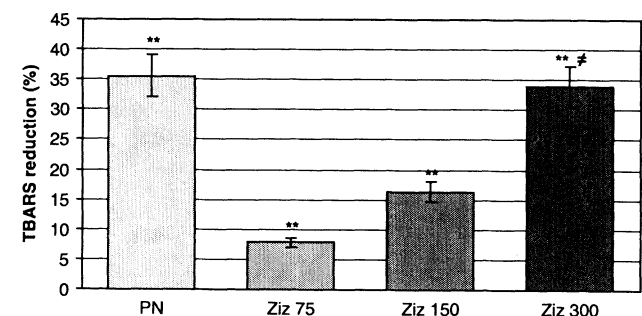


Figure 1 Effects of *Z. clinopoides* (Ziz) and prednisolone (PN) on lipid peroxidation level (TBARS) in acetic acid-induced colitis. Each value represents mean $\pm$ SEM percentage of reduction of acetic acid-induced TBARS elevation by treated compounds. The mean $\pm$ SEM of TBARS level in normal and control groups were 3.14 ± 0.31 and 5.11 ± 0.50 ($\mu\text{mol/g}$ colon), respectively. **Indicates that the difference between control and treated groups is significant at *P* <0.01. #Indicates that effects of prednisolone and *Z. clinopoides*-treated groups is not different (*P* >0.01).

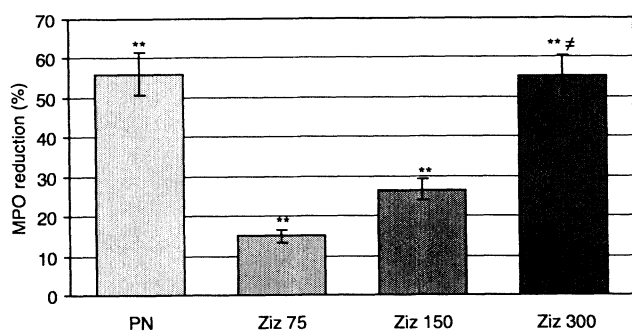


Figure 2 Effects of *Z. clinopoides* (Ziz) and prednisolone (PN) on myeloperoxidase (MPO) activity in acetic acid-induced colitis. Each value represents mean $\pm$ SEM percentage of reduction of acetic acid-induced MPO elevation by treated compounds. The mean $\pm$ SEM of MPO activity in normal and control groups were 2.89 ± 0.17 and 4.76 ± 0.15 (U/g colon), respectively. **Indicates that the difference between control and treated groups is significant at $P < 0.01$. #Indicates that the effects of prednisolone and *Z. clinopoides*-treated groups is not different ($P > 0.01$).

reduces MPO activity and TBARS concentrations; both established indicators of toxic oxidative stress and markers of colitis.^{5-9,16} Interestingly, *Z. clinopoides*, especially at a dose of 300 mg/kg, was very comparable to prednisolone, which showed significant protection against acetic acid-induced colitis proved by biochemical, macroscopic and microscopic examinations. The acetic acid-induced colitis model in mice is similar to human ulcerative colitis in terms of histologic features. It affects the distal colon portion with non-transmural inflammation, massive necrosis of mucosal and submucosal layers, mucosal edema, neutrophil infiltration of the mucosa and submucosal ulceration. The protonated form of the acid liberates protons within the intracellular space, causing a massive intracellular acidification, resulting in massive epithelial damage. Inflammation is the pathogenesis of IBD and several pathways are associated with inflammatory response in IBD. The inflammatory response initiated by acetic acid includes activation of cyclooxygenase and lipo-oxygenase pathways.^{27,29} Prednisolone as an anti-inflammatory agent decreases the recruitment of macrophages in the affected area and suppresses the synthesis of many inflammatory mediators, eg, production of IL-1 from monocytes, production of IL-2 and tumor necrosis factor from lymphocytes. Prednisolone also inhibits the enzyme phospholipase A₂ and thus decrease the availability of prostaglandins and leukotrienes.³⁰ Therefore, the first concept that may come to mind is that *Z. clinopoides* possesses an inhibitory effect on the synthesis or release of these inflammatory mediators. Supporting this idea, *Z. clinopoides* is known for its anti-diarrheic and anti-pyretic effects.^{31,33} An ethnomedical use of *Z. taurica* ssp. *taurica* for the

treatment of gastrointestinal disorders has been described.³³ Literature survey also showed that the oil of *Ziziphora* species is rich in pulegone.^{35,36} The chemical composition of the essential oil of *Z. clinopoides* from Turkey and Iran has been studied by GLC,³⁵ GC and GC-MS, respectively.³⁶ The major constituents of the oil were pulegone, pipertone, *p*-menth-3-en-8-ol, thymol, 1,8-cineole, neomenthol, menthone and isomenthol.³¹⁻³⁶ The molecular mechanism of radical scavenging activity of polar extracts from *Z. clinopoides* could be attributed to the presence of polyphenolic compounds.³⁶ It has been shown that polyphenolic compounds were responsible for radical scavenging activity in the Lamiaceae family due to the ease of their hydrogen atom donation to active free radical.³⁷ Oxidative stress is believed to play a key role in the pathogenesis of IBD-related intestinal damage.^{5-9,38-41} As a matter of fact, intestinal mucosal damage in IBD, including Crohn's disease (CD) and ulcerative colitis (UC), is related to increased free radical production and to a low concentration of endogenous antioxidant defense.^{42,43} Supporting this idea, the activity of MPO, which is a major enzyme in the formation of reactive oxygen species, is increased in the present colitis model. Together with the membranous NADPH oxidase, MPO is involved in the formation of reactive oxygen species and oxidation of biological material. It produces not only oxidative equivalents, but also contributes to the regulation of general response to invading micro-organisms.⁴⁴ Increased TBARS as a marker of lipid peroxidation observed in association with higher MPO activity in the present experimental colitis model again confirm the role of toxic free radicals in IBD. Flavonoids as the main constituent of *Z. clinopoides* extract are a class of plant phenolics with significant antioxidant and chelating properties. Their positive effects come from their ability to inhibit lipid peroxidation, chelate redox-active metals, and attenuating other procedure involving reactive oxygen species. From this point of view, there is great interest in the interactions between the polyphenolic substance and the intestinal mucosa, including their possible role as modulators of proliferation and apoptosis in the rapidly proliferating cells of the gastrointestinal epithelia.^{45,46} Recent studies have shown that flavonoids exert a beneficial effect in experimental colon inflammation.^{47,48} The beneficial effects of flavonoids increased by their antinociceptive potential.^{49,50} In addition, the anti-inflammatory effects of flavonoids seem to be related to a decrease of neutrophils infiltration, absence of upregulation of IL-1 β and decrease of prostaglandin production in

colon mucosa.⁵¹ In supporting this idea, there is evidence that flavonoids have anti-phosphodiesterase activity and, thus, could elevate intracellular levels of cyclic nucleotides.⁵² Recent studies indicate that both cAMP and cGMP can diminish oxidative stress in many biological systems and diseases.^{53–58} On the other hand, the antioxidant activity of the *Z. clinopoides* extracts and the essential oil were evaluated in a series of *in vitro* tests.^{35,36} Therefore, the beneficial effects of *Z. clinopoides* in IBD is mainly due to its strong antioxidant potential.

The *Z. clinopoides* oil and MeOH extracts exhibited strong antibacterial activity against seven Gram-(+) or Gram-(–) bacteria. In addition, the antibacterial activity of the oil may, in part, be associated with the presence of thymol and pulegones.³⁶ Thus, the antibacterial activity of *Z. clinopoides* can be

another mechanism of action for its beneficial effects in IBD. The molecular mechanism of the antibacterial activity of thymol has already been investigated and shown to cause disintegration of cell membrane.⁵⁹ Pulegone has a similar structure to carvone, which has been shown to affect cell membrane by dissipation of pH gradient and membrane potential of the cell.⁶⁰ Taken collectively, the present data provide very interesting evidence of the benefit of *Z. clinopoides* in an experimental model of toxic IBD. These effects back the anti-oxidative stress, anti-microbial, and anti-inflammatory potentials of *Z. clinopoides*. Regarding the resemblance of acetic acid-induced colitis to human UC,⁶¹ it can be concluded that *Z. clinopoides* extract may be useful in treating UC in humans. Proper clinical investigations should be carried out to confirm the same activity in human disease.

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